

Probability and Statistics – BSDSF23 (Morning)

Total Time: 50 minutes

Quiz 4

Maximum Marks: 50

Instructor: Dr. Faisal Bukhari

Obtained Marks: _____

Name: _____

Roll No. _____

1. According to a genetics theory, a certain cross of guinea pigs will result in red, black, and white offspring in the ratio 8:4:4. Find the probability that among 8 offspring, 5 will be red, 2 black, and 1 white. (10)

$n = 8$ (total number of offspring)

$x_1 = 5$ (number of red offspring)

$x_2 = 2$ (number of black offspring)

$x_3 = 1$ (number of white offspring)

$p_1 = \frac{8}{16}$ (probability of a red offspring)

$p_2 = \frac{4}{16}$ (probability of a black offspring)

$p_3 = \frac{4}{16}$ (probability of a white offspring)

Using Multinomial Distribution:

$$P(5 \text{ red}, 2 \text{ black}, 1 \text{ white}) = \frac{8!}{(5! \times 2! \times 1!)} \times \left(\frac{8}{16}\right)^5 \times \left(\frac{4}{16}\right)^2 \times \left(\frac{4}{16}\right)^1$$

$$= \frac{21}{256}$$

2. Changes in airport procedures require considerable planning. Arrival rates of aircraft are important factors that must be taken into account. Suppose small aircraft arrive at a certain airport, according to a Poisson process, at the rate of 6 per hour. Thus, the Poisson parameter for arrivals over a period of hours is $\mu = 6t$. (15)

a) What is the probability that exactly 4 small aircraft arrive during a 1-hour period?

Arrival rate (λ) = 6 per hour

Time period (t) = 1 hour

Poisson parameter (μ) = $\lambda t = 6 * 1 = 6$

Using Poisson Distribution:

$$P(X = 4) = \frac{(e^{-6} * 6^4)}{4!}$$

$$= 0.1339.$$

b) What is the probability that at least 4 arrive during a 1-hour period?

$$P(X \geq 4) = 1 - P(X < 4)$$

$$= 1 - 0.1512 = 0.8488$$

3. Suppose the probability that any given person will believe a tale about the transgressions of a famous actress is 0.8. What is the probability that (15)

a) the sixth person to hear this tale is the fourth one to believe it?

$p = 0.8$

$k = 4$

Using the negative binomial distribution,

$$b^*(x; k, p) = {}_{x-1}C_{k-1} p^k q^{x-k}$$

$$b^*(6; 4, 0.8) = {}_5C_3 (0.8)^4 (0.2)^2$$

$$= 0.1638.$$

b) the third person to hear this tale is the first one to believe it?

$$\begin{aligned} g(x; p) &= p q^{x-1} \\ g(3; 0.8) &= 0.8 (0.2)^2 \\ &= 0.032 \end{aligned}$$

4. A foreign student club lists as its members 2 Canadians, 3 Japanese, 5 Italians, and 2 Germans. If a committee of 4 is selected at random, find the probability that all nationalities are represented. (10)

The extension of the hypergeometric distribution gives a probability

$$\frac{\binom{2}{1} \binom{3}{1} \binom{5}{1} \binom{2}{1}}{\binom{12}{4}} = \frac{4}{33}$$